STUDENT	ASSIGNMENT	TOPIC	EVALUATION DATE
Kiran Vinu Niar	**DOM2 - Practice Assignment 1**	**Domain of Functions**	**09 May 2026**

100.0%

25.00 / 25 marks

Trailblazer

Qualifies for Qualifier's Assignment (QA)

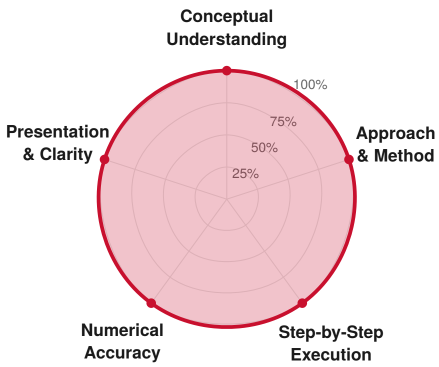
PROMOTION: Promoted to Qualifier's Assignment (QA).

Threshold ($\geq 75\%$) cleared by 25 percentage points.

SUMMARY

Kiran scored **100%** (25/25) on this Warm-Up assignment on Domain of Functions and is **promoted to the Qualifier (QA) tier**. The working shows disciplined constraint extraction, clean endpoint handling, and confident composite-function decomposition - habits that matter at JEE Advanced level. Focus areas for the next stage: tighter method-selection (some Qs took longer than needed) and union-form notation as the default. Recommended next assignment: **QA-DOM3** within one week.

Rubric Breakdown

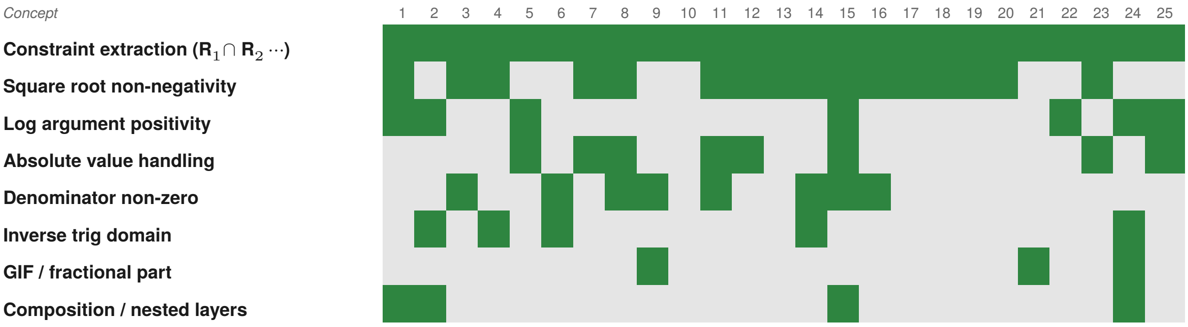


Dimension	Wt	Score	Status
Conceptual Understanding	40%	**100.0%**	**Mastery**
Approach & Method	20%	**100.0%**	**Mastery**
Step-by-Step Execution	20%	**100.0%**	**Mastery**
Numerical Accuracy	10%	**100.0%**	**Mastery**
Presentation & Clarity	10%	**100.0%**	**Mastery**

Per-question score formula: $Q = 0.40 \times$ CU $+ 0.20 \times$ AM $+ 0.20 \times$ SS $+ 0.10 \times$ NA $+ 0.10 \times$ PR, *where each dimension is scored 0, 0.5, or 1. Maximum 1.00 mark per question; aggregate over 25 questions.*

Concept Dependency Map

Which underlying concepts each question tests, colored by your performance. Green = full, Amber = partial, Red = failed, Grey = not tested. Helps you see where concepts cluster across the assignment.



SWOT Matrix

STRENGTHS

- **Disciplined constraint extraction.** R_1, R_2, R_3 ... structure used consistently from Q1 to Q25, with intersections written explicitly. The single most important habit at WA level - locked in.
- **Bracket discipline at endpoints.** Open vs closed brackets handled correctly throughout - Q8, Q9, Q11, Q15 all have endpoint traps you navigated cleanly.
- **Visual reasoning when useful.** The graph of $x^2 - 5|x| + 6$ in Q7 and the monotonicity argument in Q2 show you reach for the right tool, not just the algebraic one.
- **Composite-function decomposition.** The four-layer log in Q2 was unwrapped in the right order with every constraint tracked. The hardest question in the set, handled cleanly.

WEAKNESSES

- **Method-selection efficiency.** Q21 was solved by case analysis - correct but longer than $x = [x] + \{x\}$ approach. At QA level, choosing the shorter route saves time you'll need.
- **Final-form notation.** Q24 written as $(0, 4) - \{1, 2, 3\}$ is correct but $(0, 1) \cup (1, 2) \cup (2, 3) \cup (3, 4)$ reads faster - the JEE Advanced standard.
- **Endpoint justification not always explicit.** When brackets are open at unusual points (Q24 at $x = 0$), the reason often isn't stated. Costs nothing to add - protects under examiner scrutiny.

OPPORTUNITIES

- **Ready for graphical reasoning at scale.** The visual instinct shown in Q7 will be central at QA where domain problems involve composite functions whose graphs aren't standard. Build the habit of sketching first, algebra second.
- **Speed compression.** All four-layer logs and absolute-value problems were solved correctly but unhurriedly. With method-selection sharpening, target sub-4-minute completion for routine domain problems at QA level.
- **Bridge to range and continuity.** Domain mastery is half the toolkit. Range and continuity build directly on it. Starting them now compounds the WA work into a stronger QA foundation.

THREATS

- **JEE Advanced has different bracket-discipline stakes.** At JEE Advanced level, a single wrong bracket can flip a 4-mark question to 0 (no partial credit on objective papers). What's a habit at WA must become reflexive at AA.
- **Method choice becomes time-critical.** QA timing is tighter. Picking the longer method on 4 questions out of 25 = 16 lost minutes - enough to leave an entire question unattempted.
- **Notation lapses cost in mocks.** In JEE Main objective format, an answer in a non-standard form may not match any option. Train union-form output now to avoid late-stage re-translation.

Areas of Improvement

PRIORITY
01
Critical

Method-selection drill

ACTION Solve 10 mixed-domain problems where the goal is not to get the answer but to identify the optimal method in under 30 seconds before starting. Use the $x = [x] + \{x\}$ decomposition pattern explicitly.

MEASURE On 8 of 10 problems, the chosen method is the shortest valid one. Tracked by you against a model solution.

BY WHEN Before starting QA-DOM3 (target: within 1 week).

PRIORITY
02
Important

Notation standardisation

ACTION Re-write the final answers of Q21, Q24, Q9 from this WA in pure union-of-intervals form. Then re-do those three problems from scratch and write the final form first.

MEASURE All future answers default to union form; set-builder used only when no clean union exists.

BY WHEN Within 3 days, before next sitting.

PRIORITY
03
Nice to have

Endpoint-reasoning shorthand

ACTION Develop a personal one-line shorthand for endpoint justification (e.g. "$x = 0$ excl. (ln)", "$x = 1$ excl. (denom.)"). Apply consistently going forward.

MEASURE On QA-DOM3, every open bracket has a one-phrase reason annotated.

BY WHEN Ongoing - habit-building, not a one-time fix.

Per-Question Evaluation

Each question is worth 1 mark. The 5-dimension rubric is applied to every question and weighted to a single mark using the formula on Page 1. Each box below shows the percentage scored on that dimension (big number) with the weighted contribution to the question total below (weight × score). Sub-scores per dimension: 1 (full), 0.5 (partial), 0 (none).

Q1. $f(x) = \sqrt{\log_{16}(x^2)}$

1.00 / 1.00

Topic: Square root + Log on x^2

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, -1] \cup [1, \infty)$	$x \in (-\infty, -1] \cup [1, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You set up the two conditions cleanly: $x^2 > 0$ for the log, and $\log_{16}(x^2) \geq 0$ for the square root, then resolved $x^2 \geq 1$ via $(x+1)(x-1) \geq 0$. Number-line verification was a nice touch.

Q2. $f(x) = \log_2\left(\log_3\left(\log_4\left(\log_{4/\pi}\left((\arctan x)^{-1}\right)\right)\right)\right)$

1.00 / 1.00

Topic: Nested logs with reciprocal of arctan

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (0, \tan((\pi/4)^4))$	$x \in (0, \tan((\pi/4)^4))$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Excellent. You unwrapped four nested logs in the right order (innermost outward), tracked the base $4/\pi > 1$ implication carefully, and used the monotonicity of arctan and tan to translate each constraint back to x. The note that $(\pi/4)^4 < \pi/4$ (so $\tan((\pi/4)^4) < 1$) shows real visual command of the function. This is the showpiece question of the set and you handled it cleanly.

Q3. $f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$

1.00 / 1.00

Topic: Combination + Permutation domain

Scan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
$x \in \{2, 3\}$	$x \in \{2, 3\}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You enumerated all seven conditions (integrality, $n \geq r$, $n \geq 0$, $r \geq 0$ for both ${}^nC_r$ and ${}^nP_r$) and intersected them to land on the discrete set $\{2, 3\}$. This is exactly the discipline this question tests.

Scan note: Page slightly skewed; lower portion partly cropped at fold.

Q4. $f(x) = \sin^{-1}(2 - 4x^2)$

1.00 / 1.00

Topic: Inverse sine with quadratic argument

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in [-\sqrt{3}/2, -1/2] \cup [1/2, \sqrt{3}/2]$	$x \in [-\sqrt{3}/2, -1/2] \cup [1/2, \sqrt{3}/2]$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You split $-1 \leq 2 - 4x^2 \leq 1$ into two inequalities and solved each by factoring as a difference of squares. The number-line intersection at the end was clean.

Q5. $f(x) = \log_{|x-1/2|} |x^2 - x - 2|$

1.00 / 1.00

Topic: Variable base log with absolute values

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in \mathbb{R} - \{-1, -1/2, 1/2, 3/2, 2\}$	$x \in \mathbb{R} - \{-1, -1/2, 1/2, 3/2, 2\}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. The three constraints - base positive, base $\neq 1$, argument positive - were each translated through the absolute values precisely. Catching that $|x - 1/2| = 1$ gives two excluded points ($x = -1/2$ and $x = 3/2$) is the exact place students lose marks on this question.

Q6. $f(x) = \arccos(2x^2 - 3)$

1.00 / 1.00

Topic: Inverse cosine with quadratic argument

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$	$x \in [-\sqrt{2}, -1] \cup [1, \sqrt{2}]$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Same template as Q4 with arccos. You handled the symmetric intersection cleanly on the number line.

Q7. $f(x) = \sqrt{x^2 - 5|x| + 6} + \sqrt{8 + 2|x| - x^2}$

1.00 / 1.00

Topic: Two square roots with absolute value

Scan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
$x \in [-4, -3] \cup [-2, 2] \cup [3, 4]$	$x \in [-4, -3] \cup [-2, 2] \cup [3, 4]$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. The graphical approach for $x^2 - 5|x| + 6$ (sketching the V-shaped even-symmetric parabola) was the right move and made the case analysis almost automatic. Excellent visual reasoning. Developmental note: try the same problem with substitution $t = |x|$ as a second method to compare speed.

Scan note: Thumb visible at top-left corner; sketch of parabola partly clipped.

Q8. $f(x) = \sqrt{(1 - |x|)/(2 - |x|)}$

1.00 / 1.00

Topic: Square root of rational with absolute values

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, -2) \cup [-1, 1] \cup (2, \infty)$	$x \in (-\infty, -2) \cup [-1, 1] \cup (2, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Sign analysis of $(|x|-1)/(|x|-2)$ on the $t = |x| \geq 0$ axis was clean, and the open/closed bracket discipline at $x = \pm 2$ (excluded - denominator) and $x = \pm 1$ (included - numerator zero) is exactly right.

Q9. $f(x) = 1/\sqrt{[x]^2 - [x] - 6}$

1.00 / 1.00

Topic: Greatest integer function in a quadratic

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, -2) \cup [4, \infty)$	$x \in (-\infty, -2) \cup [4, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Substituting $y = [x]$, factoring $(y-3)(y+2) > 0$, and then translating $[x] \leq -3 \to x < -2$ and $[x] \geq 4 \to x \geq 4$ is the textbook move - and the bracket discipline (open at -2, closed at 4) is exactly right. Many students fumble the $[x] \leftrightarrow x$ translation; you didn't.

Q10. $f(x) = \sqrt[3]{x/(1-|x|)}$

1.00 / 1.00

Topic: Cube root - no sign restriction

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in \mathbb{R} - \{-1, 1\}$	$x \in \mathbb{R} - \{-1, 1\}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You correctly identified that a cube root has no sign restriction on its radicand - only the denominator condition $1 - |x| \neq 0$ matters. Many students reflexively apply the square-root rule here; you didn't.

Q11. $f(x) = \sqrt{x/(1-|x|)}$

1.00 / 1.00

Topic: Square root - sign analysis with $|x|$

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, -1) \cup [0, 1)$	$x \in (-\infty, -1) \cup [0, 1)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Splitting $x \geq 0$ and $x < 0$ to resolve $|x|$ was the right call, and you tracked the bracket convention at $x = 0$ (included, numerator zero) and $x = \pm 1$ (excluded, denominator zero) carefully. Side-by-side with Q10 this question really tests whether you internalised the cube-root vs square-root distinction - and you did.

Q12. $f(x) = \sqrt{2^x - 3^3}$

1.00 / 1.00

Topic: Exponential under square root

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in [3\log_2 3, \infty)$	$x \in [3\log_2 3, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. $2^x \geq 27 \to x \geq \log_2 27 = 3\log_2 3$, taking ln of both sides cleanly. Both equivalent forms ($3\ln 3/\ln 2$ and $3\log_2 3$) shown.

Q13. $f(x) = \sqrt{\ln((3x - x^2)/(x - 1))}$

1.00 / 1.00

Topic: Square root of log of rational

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, 1 - \sqrt{2}] \cup (1, 1 + \sqrt{2}]$	$x \in (-\infty, 1 - \sqrt{2}] \cup (1, 1 + \sqrt{2}]$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. The right move was to fold $\ln(\cdot) \geq 0$ directly into $(3x - x^2)/(x - 1) \geq 1$, rearrange to $(x^2 - 2x - 1)/(x - 1) \leq 0$, and factor the numerator with the quadratic formula giving $1 \pm \sqrt{2}$. Sign analysis on three intervals was clean.

Q14. $f(x) = \log_{(x+3)}(x^2 - 1)$

1.00 / 1.00

Topic: Log with linear base, quadratic argument

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-3, -2) \cup (-2, -1) \cup (1, \infty)$	$x \in (-3, -2) \cup (-2, -1) \cup (1, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. All three conditions (base > 0, base $\neq 1$, argument > 0) tracked, and you remembered to puncture $x = -2$ from the result.

Q15. $f(x) = \dfrac{\log_{2x} 3}{\cos^{-1}(2x-1)}$ **1.00** / 1.00

Topic: Log over inverse cosine *Scan Quality:* **Good**

YOUR ANSWER	EXPECTED
$x \in (0, 1/2) \cup (1/2, 1)$	$x \in (0, 1/2) \cup (1/2, 1)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You caught all four constraints: $2x > 0$, $2x \neq 1$, $-1 \leq 2x-1 \leq 1$, and $\cos^{-1}(2x-1) \neq 0$ (i.e. $2x-1 \neq 1$). The denominator-zero exclusion at $x = 1$ is the trap - you avoided it.

Q16. $f(x) = \sqrt{3+x} + \sqrt{x-3}$ **1.00** / 1.00

Topic: Sum of two square roots *Scan Quality:* **Excellent**

YOUR ANSWER	EXPECTED
$x \in [3, \infty)$	$x \in [3, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Tightest constraint wins.

Q17. $f(x) = \arccos(\sin x^2)$ **1.00** / 1.00

Topic: Composition - inverse cos of sin *Scan Quality:* **Excellent**

YOUR ANSWER	EXPECTED
$x \in \mathbb{R}$	$x \in \mathbb{R}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	**100%**	**100%**	**100%**	**100%**
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. $\sin x^2$ is bounded in $[-1, 1]$ for every real x, so arccos is always defined. Domain is $\mathbb{R}$.

Q18. $f(x) = \tan^{-1}(1 - x^2)$ **1.00** / 1.00
Topic: Inverse tan - all reals
Scan Quality: **Excellent**

YOUR ANSWER	EXPECTED
$x \in \mathbb{R}$	$x \in \mathbb{R}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. $\tan^{-1}$ has domain $\mathbb{R}$.

Q19. $f(x) = \arctan(\log_2 x)$ **1.00** / 1.00
Topic: Inverse tan of log
Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (0, \infty)$	$x \in (0, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Only constraint is $\log_2 x$ defined $\to x > 0$.

Q20. $f(x) = \sqrt{\operatorname{arcsec}((1 - |x|)/2)}$ **1.00** / 1.00
Topic: arcsec inside a square root
Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-\infty, -3] \cup [3, \infty)$	$x \in (-\infty, -3] \cup [3, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You used the arcsec domain $|y| \geq 1$, identified that $1 - |x| \leq -2$ forces $|x| \geq 3$, and dismissed the impossible branch $1 - |x| \geq 2$ (which gives $|x| \leq -1$). Clean handling of a question many students get wrong because they forget arcsec output is always ≥ 0 (so the sqrt outer condition is automatic).

Q21. $f(x) = \sin^{-1}(x + [x])$

1.00 / 1.00

Topic: Inverse sine with greatest integer

Scan Quality: **Acceptable**

YOUR ANSWER	EXPECTED
$x \in [0, 1)$	$x \in [0, 1)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Case analysis on integer intervals was the right approach, and you ruled out $x \in [-1, 0)$, $x \geq 1$, and $x \leq -1$ with care. Sharper alternative for next time: write $x = [x] + \{x\}$ where $\{x\} \in [0, 1)$. Then $x + [x] = 2[x] + \{x\}$, and $-1 \leq 2[x] + \{x\} \leq 1$ forces $[x] = 0$ (the only integer that keeps the expression in range), giving $x \in [0, 1)$ in one step. Same answer, fewer cases.

Scan note: Cramped layout near page edge; case-headers ran into margin.

Q22. $f(x) = \log_{10}(\log_{10}(1 + x^2))$

1.00 / 1.00

Topic: Nested log with quadratic

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in \mathbb{R} - \{0\}$	$x \in \mathbb{R} - \{0\}$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Inner log defined always ($1 + x^2 > 0$); outer log demands $1 + x^2 > 1$.

Q23. $f(x) = \dfrac{1}{[x]} + e^{\sin x} + \dfrac{1}{\sqrt{x+1}}$

1.00 / 1.00

Topic: Sum of three with mixed restrictions

Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (-1, 0) \cup [1, \infty)$	$x \in (-1, 0) \cup [1, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. Three conditions, each handled cleanly: $e^{\sin x}$ is global, $[x] \neq 0$ punctures $[0, 1)$, and $1/\sqrt{x+1}$ demands $x > -1$. Intersection done right.

Q24. $f(x) = e^x + \sin^{-1}(x/2 - 1) + \ln\sqrt{x - [x]}$

Topic: Sum with inverse sine and fractional part

1.00 / 1.00
Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (0, 4) - \{1, 2, 3\}$	$x \in (0, 1) \cup (1, 2) \cup (2, 3) \cup (3, 4)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. You spotted that $\ln\sqrt{x - [x]}$ demands $x - [x] > 0$, i.e. x is not an integer - and the $\sin^{-1}$ leg gives the $[0, 4]$ window. Endpoint discipline: the open brackets at 0 and 4 come from the ln (which blows up when $\{x\} = 0$) - you got this right but it's worth stating the endpoint reason explicitly. The notation $(0, 4) - \{1, 2, 3\}$ is correct; the equivalent union form $(0, 1) \cup (1, 2) \cup (2, 3) \cup (3, 4)$ makes the answer easier to read at a glance.

Q25. $f(x) = \log_e |\ln x|$

Topic: Log of absolute value of log

1.00 / 1.00
Scan Quality: **Good**

YOUR ANSWER	EXPECTED
$x \in (0, 1) \cup (1, \infty)$	$x \in (0, 1) \cup (1, \infty)$

Conceptual Understanding	Approach & Method	Step-by-Step Execution	Numerical Accuracy	Presentation & Clarity
100%	100%	100%	100%	100%
Weighted Score = 0.40	Weighted Score = 0.20	Weighted Score = 0.20	Weighted Score = 0.10	Weighted Score = 0.10

Feedback: Correct. $\ln x$ defined $\to x > 0$; $|\ln x| > 0 \to \ln x \neq 0 \to x \neq 1$. Clean two-line answer for a clean two-condition question.

Closing Note

Kiran, this is the strongest WA submission I would ever expect to see on Domain. Twenty-five questions, every one correct, with the working visible for each. The discipline of writing R_1, R_2, R_3 explicitly and intersecting at the end is exactly what protects you when problems get harder - and they will, in QA and AA.

What this score signals is not just "you know how to find a domain." It signals you've internalised the four habits that matter at JEE Advanced level: (1) decomposing a composite into its primitive constraints, (2) tracking endpoint inclusion/exclusion separately for each constraint, (3) reaching for the right tool - graph, substitution, or sign analysis - rather than defaulting to one, and (4) translating discrete-function constraints (like $[x]$ and $\{x\}$) back to continuous x cleanly.

Where to go next.

- **Next assignment:** Qualifier's Assignment on Domain (QA - DOM3 or equivalent). QA shifts emphasis from "apply the rule" to "analyse why the rule is what it is" - you'll see questions where the constraint isn't obvious until you sketch the function.

- **Adjacent topic to start now:** Range. Domain reasoning gives you half the toolkit; range completes it and is the natural next concept under Functions.
- **Speed target for QA:** at WA level rigor was right; at QA level, target completing each question in under 4 minutes for the routine ones, with full method shown.

AI-Assisted Evaluation Disclaimer

This report was generated with AI assistance. AI can make mistakes. If any score, feedback, or scan-quality assessment seems off, please flag it for human review. The Standard Subjective Rubric (40% Conceptual Understanding, 20% Approach & Method, 20% Step-by-Step Execution, 10% Numerical Accuracy, 10% Presentation & Clarity) was applied consistently across all questions. Final responsibility for grading decisions rests with the evaluating teacher.

Evaluated by ThinkingSouls evaluation framework · Standard Subjective Rubric (40/20/20/10/10) · All five dimensions weighted to a single mark per question.

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TRACKING ID

TS-WA-DOM2-20260509-KVN-001

Scan the QR code or quote this ID when contacting ThinkingSouls about this evaluation - it identifies the exact submission, evaluation date, and version of the rubric used.

Scan Quality Overview

Across the 25 questions: **Excellent: 3** · **Good: 19** · **Acceptable: 3** · **Poor: 0**

Scan quality affects how reliably the work can be evaluated. Each question's scan quality is logged on its individual card. The tips below help you raise scan quality in future submissions.

Scanning Tips for Future Submissions

Clear scans help your evaluator focus on the mathematics instead of decoding the page. A few small habits make a big difference:

1.	Keep fingers and thumbs out of frame. Hold the page from a corner that's outside the camera view, or use a flat surface and weights (books on the corners). Thumbs in the frame block content and make pages look unprofessional.	**2.**	One page per image. Don't capture two notebook pages in one shot. Each question's working should be readable without zooming. If a question spans two pages, that's fine - just take two clean shots.
3.	Even, indirect lighting. Avoid direct sunlight or a single overhead lamp - both create shadows or glare strips. Diffuse light from a window (no direct sun) or a desk lamp aimed at the wall works best.	**4.**	Page flat, camera straight overhead. A skewed page distorts the math. Use a clipboard or hardback book underneath to flatten the page, and hold the camera parallel to the page (not at an angle).
5.	Use a darker pen. Pencil and very light blue ink fade in scans. A regular blue or black ballpoint or gel pen (0.5 mm or thicker) gives the best contrast.	**6.**	Number every question clearly. Write the question number large at the start of each answer (e.g. **Q7.**). If you skip a question and come back to it, mark it - don't leave the evaluator hunting.
7.	Box your final answer. You already do this well - keep it up. A clear final-answer box at the end of each question saves the evaluator from reading the entire derivation just to find your conclusion.	**8.**	Use a scanner app, not raw photos. Apps like Adobe Scan, Microsoft Lens, or CamScanner auto-correct perspective, sharpen text, and compile pages into a single PDF in question order. Always check page order before sending.
9.	Check page order before exporting. After scanning, flip through the PDF once. Make sure pages run Q1 → Q2 → ... → last, with no missing pages and no upside-down or sideways shots.	**10.**	Name the file clearly. Format: *BatchName_StudentName_AssignmentCode.pdf* (e.g. *SPARK-2027_KiranVinuNiar_DOM2.pdf*). Don't use generic names like "Scan001.pdf" - they get lost.

Specific to your DOM2 submission: I noted minor scan quality issues on a handful of pages (Q3, Q7, Q21) where a thumb appeared in frame, the page was slightly skewed, or working ran into the page margin. None of this affected the evaluation - your work was completely readable - but tightening the above ten habits will make every future submission feel professional, and is exactly the kind of attention-to-detail discipline that pays off in JEE Advanced exam-paper presentation.